

A Modular Approach to Conversational Object Grounding in 3D Scenes Using VLMs and Prompt Chaining

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Data: https://huggingface.co/datasets/skeizer/conv_obj_ground

Abstract

We introduce Conversational Object Grounding, a new task in which an agent must identify a target object in a complex 3D scene by engaging in natural language dialogue with a user who knows the target. It extends multimodal reference resolution on a single referring expression to conversations over multiple turns, combining it with clarification question generation and answer understanding to resolve ambiguities. For this task, we propose a zero-shot modular approach that combines the reference resolution, object captioning, and dialogue capabilities of foundation models, using prompt chaining in conjunction with an image-grounded scene graph. Evaluations on a new benchmark constructed with 3D scenes from ScanNet show that the proposed modular approach outperforms end-to-end baselines in terms of both grounding performance (up to 13% improvement) and image processing cost (up to 63% reduction).

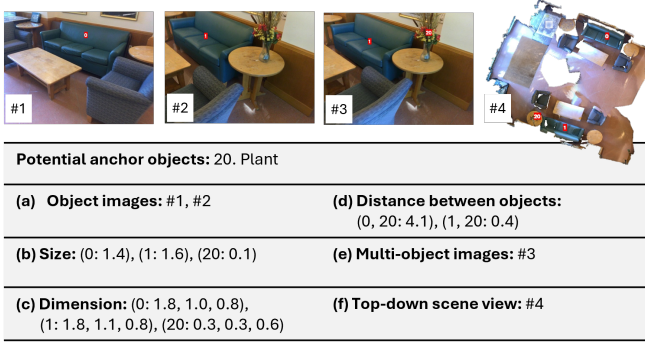
1 Introduction

To operate effectively in a physical environment, embodied AI systems (Liu et al., 2025) require perception, reasoning, and planning capabilities, and learn from their interactions with the environment to become progressively more autonomous. As robots are increasingly being deployed in spaces shared with humans, the need for natural interaction between robots and humans is becoming ever more important. To enable this, robots can be equipped with a natural language interface (Ahn et al., 2022; Park et al., 2024; Kennington et al., 2024; Quartey et al., 2024). However, interpreting and following user instructions in the context of a physical environment is a challenging task involving several aspects such as visual perception, natural language processing, and planning (Chai et al., 2018; Qi et al., 2019; Zhang et al., 2022; Sarch et al., 2023; Farag et al., 2025).

An important aspect of interpreting natural language user instructions is *object grounding*, i.e., associating object references in the instruction with perceived objects in the physical environment (Achlioptas et al., 2020; Chen et al., 2020; Zhang et al., 2023; Kottur et al., 2021; Kottur and Moon, 2023). Most work in this area focuses on making a prediction in one shot, based on a single object description. In practice, however, a description might be ambiguous, in which case the robot would need to engage in a clarification dialogue with the user to resolve the ambiguity (see for example the dialogue on the right in Figure 1).

With that in mind, we propose a new task, *Conversational Object Grounding*, in which an agent is given a 3D scene and engages in a cooperative dialogue with a user to identify a target object in the scene, known to that user. The task goes beyond the similar ‘GuessWhat?!’ task (de Vries et al., 2017) by considering complex, multi-image scenes and allowing free-form clarification dialogue.

One might expect that powerful Vision Language Models (VLMs) such as the recent GPT variants can be used to perform conversational object grounding in a straightforward manner, i.e., in each dialogue turn using a single prompt with sufficient images to represent the scene. However, complex 3D scenes and referring expressions may require advanced reasoning based on many images for effective reference resolution and clarification, resulting in higher (image processing) costs and higher risk of failure. For other 3D scene understanding tasks (including non-conversational object grounding), alternative approaches have been developed, including fine-tuning foundation models to process 3D input (Hong et al., 2023; Huang et al., 2024a; Qi et al., 2025), or employing a symbolic representation of the environment instead of images (Armeni et al., 2019; Kim et al., 2019b; Gu et al., 2024; Fang et al., 2024; Li et al., 2025). Typically, such a representation takes the form of



	Object IDs
Agent: Which object do you have in mind?	
User: The couch.	[0, 1]
Agent: I see two couches. Do you mean the one near the plant or the one further from it?	
User: The one next to the plant.	[1]
Agent: Okay, the couch near the plant.	

Figure 1: Example conversational object grounding dialogue with predicted object IDs (on the right) and multimodal information extracted from an Image-grounded Scene Graph used by the agent (on the left).

a *scene graph*, where the nodes are objects and the edges are relations between the objects (e.g. ‘above’ or ‘behind’). Although scene graph methods enable LLM prompting, giving them open vocabulary language understanding and generation capability, VLM prompting with images can provide the agent with more information about the environment than what can be expressed through the closed set of predefined properties and relations in a symbolic scene graph.

To address the above issues, we propose a zero-shot approach to conversational object grounding that combines *prompt chaining* with *image-grounded scene graphs*. The method uses a modular structure to efficiently leverage the language understanding and generation abilities of LLMs and the multimodal reference resolution and captioning abilities of VLMs. In particular, we adopt a prompt chaining mechanism that enables the dynamic infusion of multimodal information obtained through an image-grounded scene graph. Rather than a comprehensive symbolic representation of the environment, this scene graph contains only basic symbolic information, such as the type, centroid, and dimensions of all objects in the scene, and is supplemented with an image-to-object mapping that enables dynamic selection of relevant images for specific VLM prompts. This approach enables not only a more focused use of VLMs, but also reduced image processing costs.

A new benchmark for conversational object grounding has been constructed by selecting 3D scenes from ScanNet (Dai et al., 2017) and identifying object types that have multiple instances, setting each instance as the target object for a new task. In each of the resulting tasks, the agent aims to identify the target by asking clarification questions to a user, who is given the target object ID

beforehand and can therefore answer the questions. Although the interactions are set up to be cooperative, the success of the grounding task depends on the quality of both the agent’s clarification questions and the user’s answers.

In extensive evaluations on this benchmark, we show that the proposed modular conversational object grounding method outperforms an end-to-end baseline by 13% relative in object grounding F-score clarification gain, using 63% fewer images for VLM prompting.

To sum up, we offer the following contributions:

1. **Conversational Object Grounding**, a new dialogue task for 3D object identification.
2. A **Modular zero-shot approach** to conversational object grounding using VLMs combined with prompt chaining.
3. **Evaluations on a new dataset** based on 3D scenes from ScanNet, comparing the proposed modular approach with end-to-end baselines.

2 Method

The conversational grounding task is for an agent to identify a target object in a 3D scene by engaging in a clarification dialogue with a user who knows which object is the target. In our experimental setup, a dialogue always starts with the agent asking “Which object do you have in mind?”, followed by the user answering with an object description (see Figure 1). The agent then predicts candidate objects that the user might be referring to. If this is a single candidate, it will be the final prediction and the agent closes the dialogue, confirming with an unambiguous object description; otherwise, the agent deems the object description ambiguous and generates a clarification question.

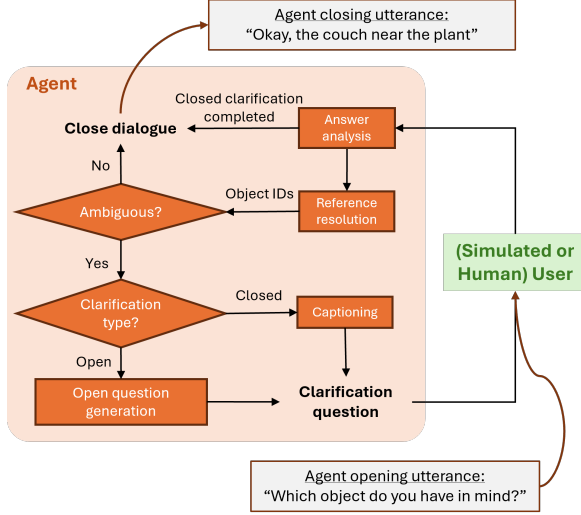


Figure 2: Conversational Grounding Agent flowchart

The key idea behind our proposed method is to use a modular structure that enables *Prompt Chaining* in conjunction with an *Image-grounded Scene Graph*. The overall structure, shown in Figure 2, enables leveraging both the language understanding and generation abilities of LLMs and the multimodal object captioning and reference resolution abilities of VLMs, using the scene graph for dynamically selecting multimodal input. Next, we describe the individual components in more detail.

2.1 Agent architecture

For the agent, **reference resolution** (Section 2.3) is the core functionality of grounding, i.e., predicting a list of object IDs given the dialogue so far (containing one or more object references) and information about the environment (including scene images). When the agent cannot identify a unique object ID as the target, it generates a clarification question. For a *closed clarification question*, e.g., “Do you mean the black chair near the door or the brown chair next to the table?”, **captioning** (Section 2.4) is used to describe the candidate objects as options for the user to choose from. An *open clarification question*, e.g., “I see three chairs. Which one do you mean?”, summarises the ambiguity and asks the user for a more specific object description. Currently, the agent generates a closed clarification question when there are two candidates and an open clarification question when there are more than two candidates. More sophisticated strategies for selecting types of clarification questions can be considered, but are left for future work.

After the user has responded with an answer, the

agent performs **answer analysis** (Section 2.5) to determine whether it can directly make a singleton prediction. If it can be inferred that the user has selected one of the options offered in a closed clarification question, the corresponding object ID is used as final prediction. Otherwise, the agent performs **reference resolution** as described above, using the answer.

To enable automatic evaluations, a **Simulated User** with similar modular architecture was developed (see Figure 5 in the Appendix), which uses similar modules for object captioning (describing the target object), reference resolution (processing closed clarification questions) and dialogue handling (question analysis and answer formulation).

2.2 Image-grounded scene graph

Where previous methods used a textual scene graph to prompt an LLM (Fang et al., 2024; Gu et al., 2024), we introduce an *Image-Grounded Scene Graph (IGSG)* to prompt a VLM. To construct a prompt for a specific VLM task, relevant information about the environment is dynamically retrieved from this graph, which may include both text and images. Our scene graph contains only basic information such as the type, location and dimensions of each object, leaving attributes like colour, and spatial relations such as ‘above’ and ‘next to’ to be inferred from images, facilitated through an image-to-object mapping (see Appendix C).

We use real-world scenes from ScanNet, in which the scene graphs are constructed based on the available *3D point cloud*, which itself is constructed from a set of images of the scene (Dai et al., 2017). The images fed to the VLM are annotated on-the-fly with IDs of relevant objects, depending on the prompting subtask. The IDs are positioned at the center pixel of the 2D segmentation masks that correspond to the object proposals from the point cloud, projected onto the image. The ID positioning is further improved using a high-quality 2D segmentation method (Kirillov et al., 2023).

2.3 Reference resolution

The process of determining which object an interlocutor is referring to happens in four stages. The **Expression Extraction** and **Information Specification** stages involve text-only LLM prompting, the **Information Retrieval** stage involves querying the scene graph, and the **Target Object Prediction** stage involves feeding selected images as well as text to a VLM. An overview of this pipeline is

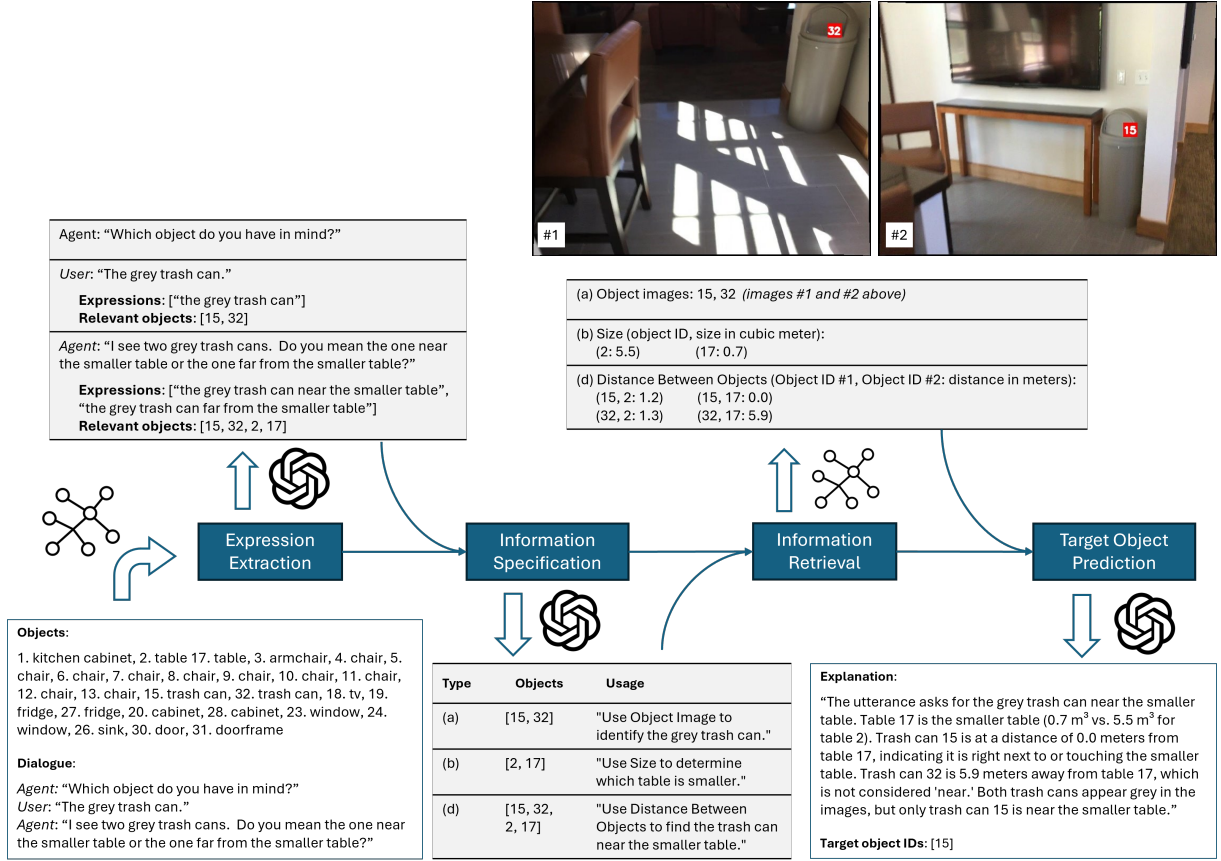


Figure 3: Reference resolution overview, indicating scene graph based and LLM/VLM based steps. The example illustrates the simulated user processing a closed clarification question, extracting the two alternative object descriptions, and then identifying the corresponding objects (in two separate iterations, the first of which is shown here).

shown in Figure 3, starting from the bottom left with a list of objects and a dialogue as input, and ending up with a prediction on the bottom right as output. Please note that this particular example describes the *simulated user* performing reference resolution to answer a closed clarification question asked by the agent.

In the **Expression Extraction** stage, an LLM is prompted to extract expressions from the dialogue that contain one or more references to objects, and for each expression list the relevant object IDs (for the prompts, see Figures 11 and 13 in the Appendix). In the example in Figure 3 the user tries to identify the alternative object IDs for the closed clarification question asked by the agent in the second turn ("I see two grey trash cans. Do you mean the one ... or the one ...?"). To that end, the two phrases describing the alternatives are extracted from the question and combined with the IDs of all objects of the mentioned types ('trash can' and 'table'). All extracted expressions are shown on the top left in Figure 3.

In the **Information Specification** stage, an LLM is prompted to determine for each of the extracted expressions with relevant object IDs which types of information are required to predict the referent object, selecting from the following types (for the prompt, see Figure 14 in the Appendix):

- (a) **object image**: provides visual details of an object, e.g., colour, shape, etcetera,
- (b) **size**: to distinguish between big and small objects,
- (c) **dimensions**: to differentiate wide and narrow, long and short, and tall and short objects,
- (d) **distance**: to indicate whether objects are near to or far from each other,
- (e) **multi-object image**: provides spatial relationships, such as 'above', 'under', 'next to', 'behind', and 'in front of',
- (f) **top-down scene view**: gives the overall scene layout.

The generated information specification for one of the extracted expressions is shown in the bottom

centre table of Figure 3, providing for each item the type of information (a-f), the relevant object IDs, and an explanation of how the information is to be used in the prediction stage.

In the **Information Retrieval** stage, the actual information according to the generated specification is extracted from the scene graph and passed on to the **Target Object Prediction** stage. In this final stage, a VLM is prompted (see Figure 15 for the prompt) to generate a target object prediction, in the form of a list of candidate object IDs, along with an explanation, as exemplified on the bottom right in Figure 3.

2.4 Object captioning

Generating object descriptions for the purpose of simulating user answers (to an open question) or constructing closed clarification questions is done in two stages: **Information Sampling** from the scene graph and **Object Description Generation** based on the sampled information; see Figure 4 (a more detailed overview is given in Figure 6).

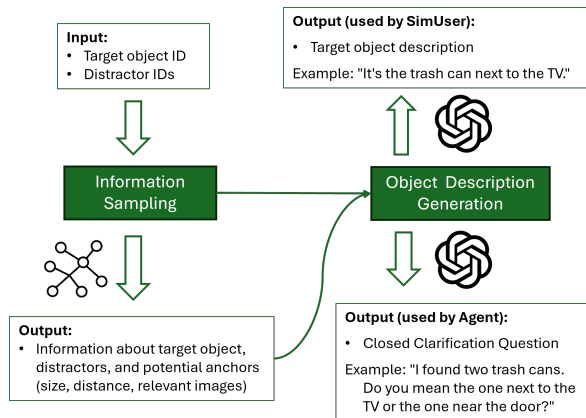


Figure 4: Captioning overview.

During **Information Sampling**, information that might be useful in describing a given target object is heuristically extracted from the scene graph. In addition to the target and its distractors (other objects of the same type), potential anchor objects are included, restricted to objects with types that have only one instance, in order to avoid confusion.

In the **Object Description Generation** stage, a VLM is prompted (see Figures 16 to 18) with the sampled scene graph information, including the selected images, to generate an object description (see the example on the top right in Figure 6) or a closed clarification question (see the example output on the bottom right in Figure 6) describing a limited number of alternative objects.

2.5 Dialogue components

The remaining components involve prompting an LLM to perform a task, providing only the dialogue so far as input. In **Question analysis**, the simulated user decides whether the agent asked an open or closed clarification question (see Figure 12). In **Answer formulation**, the simulated user generates an answer to a closed clarification question, based on reference resolution results on the question (see Figure 20). In **Answer analysis**, the agent decides if the target object can be inferred directly from the user answer (see Figure 10). Finally, in **Open question generation**, the agent generates an open clarification question (see Figure 19).

3 Experiments

To evaluate the proposed method for conversational object grounding, we have created a dataset of tasks based on scenes from ScanNet (Dai et al., 2017) and then generated grounding dialogues for each task in various settings.

3.1 Metrics

Object grounding performance is measured in terms of **Precision**, **Recall**, and **F-score** of the predicted object IDs against the ground-truth target object ID. Aiming for a high recall on ambiguous object descriptions, the clarification dialogue should help narrow down the list of candidate objects, thus improving precision. We compare performance levels **before clarification** (one-shot grounding) and **after clarification**, as well as absolute and relative **F-score clarification gains**. In addition, we report the average **number of images** included in VLM prompts, per turn and per dialogue.

3.2 Data

To create a benchmark for the conversational grounding task, we selected 65 ScanNet scenes and generated the corresponding image-grounded scene graphs based on the available 3D point cloud information. In each scene, we identified the object types that had at least two, and up to four instances, and then created a data point for each of the instances, marking it as the target object. This process resulted in a dataset of 612 tasks, each with a target object that has at least one and up to three distractors, 1.49 on average.

Although the bulk of our evaluation dialogues were generated from scratch using our LLM/VLM-based agent and simulated user, we also ran experi-

ments using object descriptions from Multi3DRefer (Zhang et al., 2023). This is a benchmark for multi-modal reference resolution in scenes from ScanNet, extending the ScanRefer benchmark (Chen et al., 2020) of human-generated single target referring expressions with zero-target and multi-target expressions, further augmented using ChatGPT. We used the multi-target object descriptions as ambiguous first user turn in our clarification dialogues, selecting one of the reference objects as the target.

3.3 Baselines

In our experiments, the proposed **Modular** method is compared with two other zero-shot baseline methods. The **End2End** method uses a single prompt to jointly predict candidate object referents and generate a response in every dialogue turn. Rather than dynamically selecting a limited set of images for each step in the modular prompt chaining approach, it uses a fixed set of images for every scene, selected such that every object (represented in the scene graph) appears at least once, in addition to a top-down view image. The **TwoStage** method is also end-to-end, insofar as it jointly predicts candidate objects and generates a response in a single VLM call. However, it uses an additional LLM call to extract the target object type from the current dialogue to enable a much more focused selection of images for the VLM call. Further details about the VLM settings can be found in Appendix D.

3.4 Experimental conditions

The simulated user was employed to generate object grounding dialogues in interaction with the agent for each task in our dataset. The comparison between the three methods was carried out in two conditions, defined by the nature of the first user turn: in the **Ambig** setting, an ambiguous description of the target object referring only to the object type was generated in template-based fashion, e.g., “it is a chair”; in the **Unambig** setting, a VLM is prompted to generate an unambiguous description of the target object (which may in practice not be unambiguous and therefore require clarification).

3.5 Results

The results in Table 1 show a consistent pattern of grounding F-score improvements achieved through clarification. When we force the first user turn (providing the initial object description) to be ambiguous (Ambig), gains of over 30% are achieved across all three agent architectures. Even when the

first user utterance is intended to uniquely describe the target object (Unambig), clarification improves grounding performance, with gains of up to 5%.

More importantly, we observe that the proposed Modular method outperforms the End2End and TwoStage baselines by 4.0% and 4.4% in F-score gain respectively. Furthermore, the Modular method is much more efficient in its use of images, selecting only 6.7 images per turn for VLM prompting, where the End2End baseline uses 17.5 images, achieving lower grounding scores. The results for the TwoStage baseline show that a basic selection mechanism can drastically reduce image processing costs compared to a more brute-force VLM prompting strategy, but this may deteriorate grounding performance. Our Modular approach provides a more sophisticated and focused way of using images, resulting in more effective grounding without using an abundance of images.

Sensitivity to VLM The above results are all obtained using gpt-4.1¹ for all VLM calls. Here, we investigate how the results of our proposed modular method are affected if we replace the VLM with a smaller, open-source, Qwen model (Qwen3-VL-30B-A3B-Instruct); see (Yang et al., 2025). The results in Table 2 show that overall, a less powerful VLM results in lower grounding performance and weaker clarification gains. More specifically, the results reflect the relative importance of the VLM in the Unambig/Ambig conditions and before/after clarification stages. When replacing the Agent VLM with the Qwen model, the drop in F-score before clarification is much more limited in the Ambig condition than in the Unambig condition (1% vs 8%), because the initial user utterance does not require multimodal reference resolution. In the Ambig condition, the VLM is much more important in the clarification stage, which is reflected in the reduced gains made when using Qwen for the Agent (+29% vs +35%).

Task complexity Our evaluation dataset consists of tasks with target objects that have at least one and up to three distractors, i.e., objects of the same type. To show how the performance of our conversational grounding method is affected by this kind of task complexity, we break down the results into the tasks with only one distractor and remaining tasks with at least two distractors. The results in Table 3 show that a task is more challenging when

¹<https://openai.com/index/gpt-4-1/>

Table 1: Evaluation results, in terms of target object prediction **Precision**, **Recall**, and **F-score**, as well as number of images included in VLM calls, using gpt-4.1 for both Agent and Simulated User throughout.

User 1st turn	Agent archit	Before clarification			After clarification			F-score Gain		Images per		Agent turns
		P	R	F	P	R	F	Abs.	Rel.	turn	dialog	
Unambig	End2End	76.0	92.0	83.2	87.7	88.5	88.1	+4.9	+5.8	17.5	21.5	1.23
	TwoStage	74.3	88.4	80.7	84.7	85.2	85.0	+4.2	+5.2	6.1	7.5	1.24
	Modular	84.1	90.2	87.0	89.7	88.7	89.2	+2.2	+2.5	6.5	7.4	1.15
Ambig	End2End	39.4	99.7	56.5	86.5	88.8	87.7	+31.2	+55.2	17.5	37.5	2.14
	TwoStage	38.9	99.5	55.9	86.2	87.3	86.7	+30.8	+55.2	6.1	12.9	2.12
	Modular	39.0	99.5	56.0	91.1	91.3	91.2	+35.2	+62.9	6.7	13.9	2.06

Table 2: Evaluation results, in terms of target object prediction **Precision**, **Recall**, and **F-score**, comparing two different agent VLMs, using the modular method throughout.

User 1st turn	Agent model	Before clarification			After clarification			F-score Gain	
		P	R	F	P	R	F	Abs.	Rel.
Unambig	GPT	84.1	90.2	87.0	89.7	88.7	89.2	+2.2	+2.5
	Qwen	77.5	80.7	79.0	81.3	80.7	81.0	+2.0	+2.5
Ambig	GPT	39.0	99.5	56.0	91.1	91.3	91.2	+35.2	+62.9
	Qwen	38.2	98.5	55.0	83.4	81.7	82.6	+27.6	+50.1

Table 3: Evaluation results for the combined Ambig and Unambig conditions, using the Qwen VLM throughout: one vs multiple distractors.

Distrs	Size	Cond	Before	After	Gain	RelGain
1	792	<i>E2E</i>	59.8	70.1	+10.3	+17.3
>1	432		39.7	59.9	+20.1	+50.6
1	792	<i>Mod</i>	71.3	88.3	+17.0	+23.9
>1	432		48.0	68.0	+19.9	+41.5

there are more distractors, but clarification seems to have a levelling effect. Before clarification, the difference in F-score is more than 20%, reduced to just over 10% after clarification.

Table 4: Ablation study results (F-scores for the combined ambig and unambig conditions and Qwen model for both Agent and SimUser).

Modality	Before	After	Gain	RelGain
Text-only	59.1	76.6	+17.5	+29.7
Multimodal	61.6	81.1	+19.5	+31.6

Ablation study An important feature of our Modular conversational grounding method is access to images for VLM prompting via an image-grounded scene graph. To analyse this further, we created a text-only version of our method which only relies on the symbolic information, without access to any images. The results in Table 4 reveal that all F-scores are notably worse in the text-only version. Hence, our method leverages additional infor-

mation in the images for improved conversational grounding performance. Note that grounding before clarification in the Ambig condition does not require any image processing, which is why the difference in F-scores is much more limited than after clarification.

Multi3DRefer descriptions If we use the multi-target object descriptions from Multi3DRefer for the first user turn (see Section 3.2), our Modular grounding method achieves effective clarification, but reaches considerably lower F-scores than in the Ambig condition (see Table 5). We believe this is largely down to a loss in recall on the initial descriptions, which are meant to be multi-target and then set up in our experiment to be ambiguous, but often interpreted by our method as unambiguous. A closer analysis revealed that the ambiguity of some of the descriptions is in fact debatable, for example when standing somewhere in a bathroom there may be two towels on the left, but when standing in front of them, only one of them is on the left.

Table 5: Results when using Multi3DRefer (M3DRef) descriptions, including F-scores relative to multi-target labels before clarification (**MT**), and to single-target labels **Before** and **After** clarification.

Cond	Size	MT	Before	After	Gain	RelGain
Ambig	608	98.3	56.0	91.2	+35.2	+62.9
M3DRef	578	76.1	50.6	67.0	+16.4	+32.4

Human user study In addition to the automatic evaluations, a preliminary experiment with human users was carried out, recruiting eight subjects from within our organisation. For each task, the subject was shown a series of images from a ScanNet scene, annotated with object IDs, and a 3D visualisation of the point cloud². They were then asked to help the agent identify a given target object through natural dialogue (see Figure 7). We randomly selected 79 instances across 45 scenes from our dataset and collected 158 dialogues in total (across the Ambig and Unambig conditions; see Appendix B), using the Modular method powered by the GPT VLM. The results in Table 6 suggest that the conversational grounding method enables effective clarification, achieving F-score gains of up to 24%.

Table 6: Human evaluation results, along with automatic evaluation results for the same data subset.

User	Condition	Before	After	Gain	RelGain
SIMUL	Ambig	53.5	90.9	+37.4	+69.9
	Unambig	88.9	91.5	+2.6	+2.9
HUMAN	Ambig	53.3	76.9	+23.6	+44.8
	Unambig	59.7	73.4	+13.7	+22.9

4 Related Work

Although clarification in dialogue has been studied extensively (Purver et al., 2001; Schlangen, 2004; Rieser and Lemon, 2006; Stoyanchev et al., 2013; Khalid et al., 2020; Benotti and Blackburn, 2021; Aliannejadi et al., 2021; Li et al., 2024; Mazzacara et al., 2024), in multimodal settings it is still an emerging topic (de Vries et al., 2017; Gervits et al., 2021; White et al., 2021; Kottur et al., 2021; Matsuzawa et al., 2023; Chiyah-Garcia et al., 2023).

Various benchmarks related to ambiguity and clarification in situated dialogue have been developed: for the IGLU competition (Kiseleva et al., 2021, 2022) the MineCraft corpus (Narayan-Chen et al., 2019) was extended with clarification questions, Madureira and Schlangen (2023) have provided annotations of instruction clarification requests (iCRs) in the CoDraw dataset (Kim et al., 2019a), and in (Kottur and Moon, 2023), multimodal reference resolution tasks were defined for both ambiguous and unambiguous object mentions, but no specific task for clarification was included. Similarly, Haber et al. (2019) offer a dataset of visu-

ally grounded dialogues and a dialogue-aware reference resolution baseline, without focusing specifically on clarification to handle ambiguities.

Prior works that offer solutions for clarification and disambiguation are all evaluated in limited settings, or focus only on part of the conversational grounding problem. Some works that consider multimodal settings focus on Y/N questions only (de Vries et al., 2017; White et al., 2021) or clarification question generation only (Gervits et al., 2021; White et al., 2021; Matsuzawa et al., 2023). Chiyah-Garcia et al (2023) focus on the understanding part of conversational reference resolution, evaluated on a static dialogue corpus, and not on the generation of clarification questions.

Many papers on 3D scene understanding report results on various tasks like reference resolution, captioning, and question answering. SeeGround (Li et al., 2025) proposes a reference resolution approach that leverages VLMs, but using query-aligned *rendered* images, rather than selected *raw* images. Zhang et al. (2025) offer a benchmark for ‘Interactive Referential Grounding’, but leave multi-turn dialogue as future work. None of the methods for object captioning (Luo et al., 2023; Huang et al., 2024b,a) have been used for clarification question generation. Although some works propose unified frameworks to perform a range of 3D comprehension tasks in isolation, our work is the first to combine several tasks into a single conversational system, providing both an agent and a simulated user to generate clarification dialogues for object grounding.

5 Conclusion

In this paper, we have introduced the task of conversational object grounding, combining reference resolution and clarification to efficiently identify target objects in complex 3D environments in interaction with a user. We have also proposed a modular approach to performing this task, combining image-grounded scene graphs with prompt chaining mechanisms to leverage LLM/VLM capabilities in zero-shot fashion. In evaluations on a custom dataset we have demonstrated that the proposed method outperforms two end-to-end baseline methods in terms of object grounding performance and image processing cost. The results show the benefits of using images in a scene graph based approach to this task, but also the importance of making targeted selections of relevant images.

²https://kaldir.vc.in.tum.de/scannet_browse/scans/scannet/querier

6 Limitations

As described in Section 3.5, we have used multi-target object descriptions from the Multi3DRefer benchmark to serve as ambiguous object references, to be resolved through clarification. Many of these descriptions are indeed ambiguous and have been labelled with multiple object IDs in that sense, but others are genuinely unambiguous and do not require clarification. This distinction has not been made in the benchmark, but is crucial in real scenarios. Our methodology does not currently support zero- or multi-target cases, but extending it to cover those should be relatively straightforward.

The clarification strategy used in this paper is focused on generating questions and processing their answers, using a simple heuristic to decide between open and closed questions, and designated prompting schemes to generate them. Furthermore, the agent does not have the option to confirm a singleton prediction with the user, allowing for recovery in case that prediction is wrong. Exploring more flexible strategies is left for future work.

In our current automatic evaluation setting, the agent and simulated user have access to the same image-grounded scene graph. In more realistic settings (as we have seen in our human user study), there may be discrepancies in perception between agent and user, or limitations in the agent’s perception itself, that have to be overcome. As we have focused more on the reasoning aspect of clarification dialogue in this paper, robustness in terms of visual perception is left for future investigations.

7 Ethical considerations

As part of our experiments, we conducted a preliminary human user study, recruiting subjects from within our organisation. The subjects were asked to engage in conversation with a chatbot by typing natural language utterances, helping the agent identify a target object in a 3D scene. Only the typed utterances were collected from the subjects, along with informal feedback about their experience with the system. All subjects gave consent to their data being used for research purposes only, to be anonymised before public release.

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A Full conversational grounding architecture

The proposed modular approach with image-grounded scene graphs and prompt chaining is used for both the agent and the simulated user that is employed to run automatic evaluations. Both have components for dialogue handling, reference resolution, and object captioning, adapted to their respective roles, as shown in Figure 5. When the agent asks a closed clarification question, the simulated user extracts the options from the question and performs reference resolution to identify the corresponding object IDs. If the target object matches one of them, an answer can be formulated accordingly; otherwise, the simulated user reverts to object captioning to describe the target object. When the agent asks an open clarification question, the simulated user goes straight into object captioning to generate an answer.

A more detailed overview of the captioning module described in Section 2.4 is shown in Figure 6. It includes an example of information sampled from the scene graph based on a target object and a distractor, as well as generated object descriptions for the purpose of describing the target object (for the simulated user) and generating a closed clarification question (for the agent).

B Human user study

For the human user study, we randomly selected 79 instances from the full test set and collected dialogues for them in two different conditions:

- **Forced ambiguity:** in their first answer, the user should only refer to the target object type, creating an ambiguous expression (roughly corresponding to the Ambig condition).
- **Unrestricted:** the user simply conveys the identity of the target object to the agent, using one or more turns of natural conversation (sitting somewhere between the Ambig and Unambig conditions).

Every subject carried out ten tasks (corresponding to ten different instances) in both conditions, each producing a dialogue for one of the 79 instances. After collecting all dialogues, including the agent’s target object predictions, precision, recall, and F-scores were calculated.

Figure 7 shows an example scenario with multiple objects of type ‘table’, only one of which is the target object, presented to the user. In the dialogue, the user first describes the target object by referring to its type only, which results in the agent predicting all three objects of type ‘table’ as candidates. After the agent asks an open clarification question, the user uses a spatial relationship with an anchor object in their answer, resulting in the agent correctly identifying object 8 as the target.

The full results of the human evaluation are shown in Table 8, confirming the effectiveness of the proposed clarification method. Furthermore, we note that higher F-scores before clarification were achieved in the Unrestricted condition, but F-scores after clarification were higher in the Forced Ambig condition, showing the benefit of clarification dialogue over one-shot object description. The results also suggest that compared to the simulated user, the human users struggled to grasp the complex 3D scenes and produce sufficiently accurate object descriptions to help the agent identify the target object. The automatic evaluation results on the human evaluation subset show similar patterns to those on the full dataset, except for the markedly lower precision in the Ambig condition, which is likely due to the higher average number of distractors in this subset (1.66 vs 1.49), see Table 7.

Table 7: Evaluation dataset statistics: number of scenes and tasks and average number of distractors per task.

Dataset	Scenes	Tasks	Distractors
Full dataset	65	612	1.49
Human eval. subset	45	79	1.66

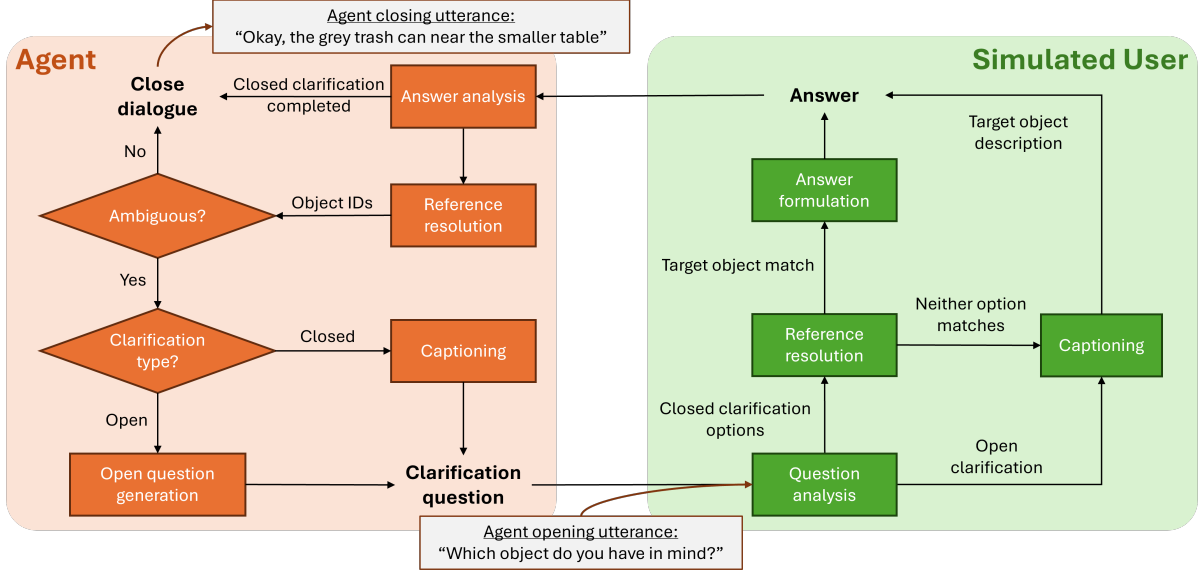


Figure 5: Conversational Grounding Agent and Simulated User flowcharts: every dialogue starts with the agent asking a general opening question (grey box at the bottom) and ends with the agent confirming the identified object (grey box at the top).

Table 8: Human evaluation results in terms of target object prediction **Precision**, **Recall**, and **F-score**, including automatic evaluation results for the corresponding data subset (all experiments using modular architecture and GPT model).

Evaluation	Condition	Before clarification			After clarification			Gain	RelGain
		P	R	F	P	R	F		
AUTOMATIC	Ambig	36.5	100	53.5	90.9	90.9	90.9	+37.4	+69.9
	Unambig	84.7	93.5	88.9	92.1	90.9	91.5	+2.6	+2.9
HUMAN	Forced ambig	36.7	97.5	53.3	77.9	75.9	76.9	+23.6	+44.8
	Unrestricted	47.7	79.7	59.7	73.4	73.4	73.4	+13.7	+22.9

C Contents of Image-Grounded Scene Graph

In our proposed method, we employ a scene graph with nodes representing objects and edges representing relations between objects, as usual. However, we only store basic information about each object and rely on the raw images of the scene to cover other object properties and relations. To facilitate this, the scene graph is grounded in the images via an image-to-object mapping, a JSON structure of 2D object centroids in each image, used to generate object ID annotations for VLM prompting.

Object properties:

- **ID**: Unique identifier for the object.
- **label**: Object type.
- **centroid**: 3D coordinates of the object’s centroid.
- **dimension**: Length, width, and height of the 3D bounding box (L/W/H).
- **size**: Overall size of the object.
- **image_indices**: List of image indices where the object appears.
- **occupancies**: List of the object’s 2D-to-3D point projection ratios in each image (number of 3D points projected onto a 2D image ÷ total number of 3D points), used to select object or multi-object images for VLM reasoning.
- **areas**: List of the object’s 2D areas in each image, used to select object or multi-object images for VLM reasoning.
- **pointcloud**: 3D point cloud of the object, used to calculate distance between objects and produce top-down scene views.

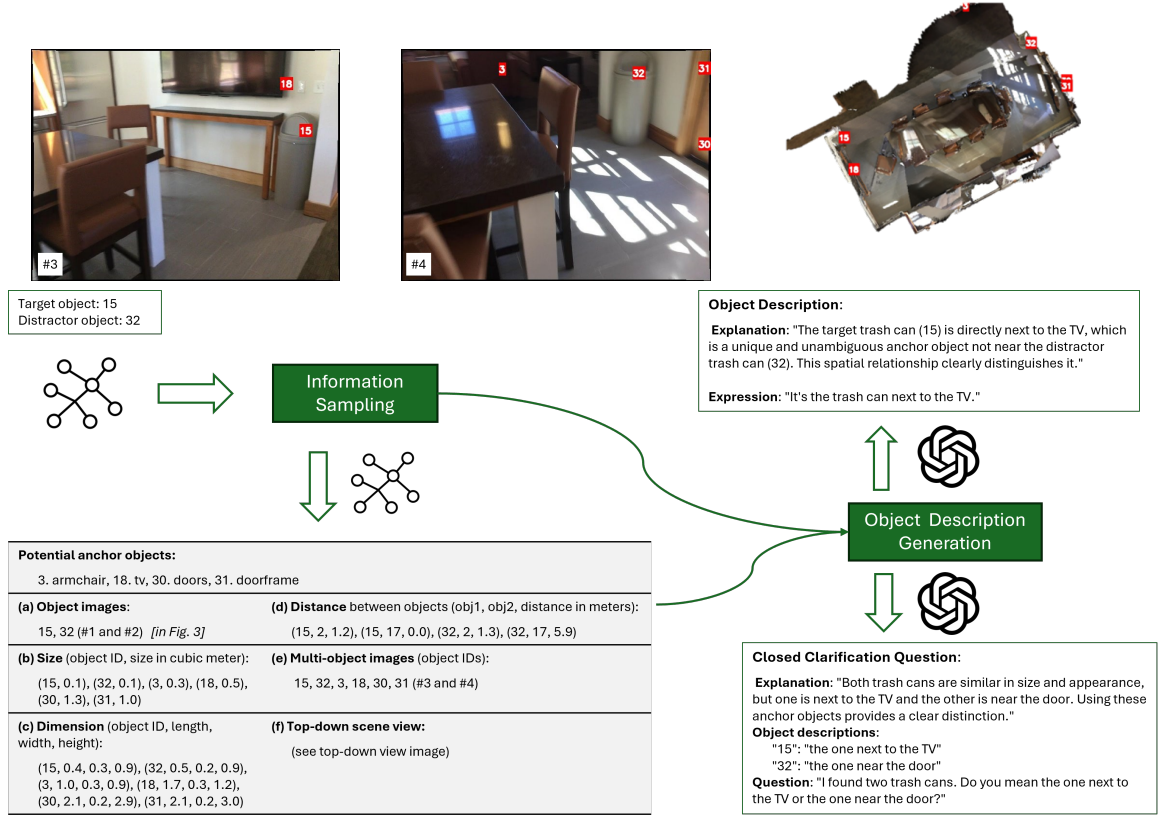


Figure 6: Captioning overview, indicating scene graph based and LLM/VLM based steps.

D Experimental details

In all experiments, the **gpt-4.1** model was accessed through the Azure OpenAI API; the **Qwen3-VL-30B-A3B-Instruct** model was run locally on 4 NVIDIA RTX A6000 GPUs. In all VLM calls, the *temperature* is set to 0.0 and the *top_p* set to 0.0001, except for the Answer Formulation, Open Question Generation, and object Description Generation steps (see Section 2), where the *temperature* and *top_p* are both set to 0.5.

E Prompts

E.1 End-to-End Baseline Agent

The End-to-end baseline agent uses a single prompt, shown in Fig. 8.

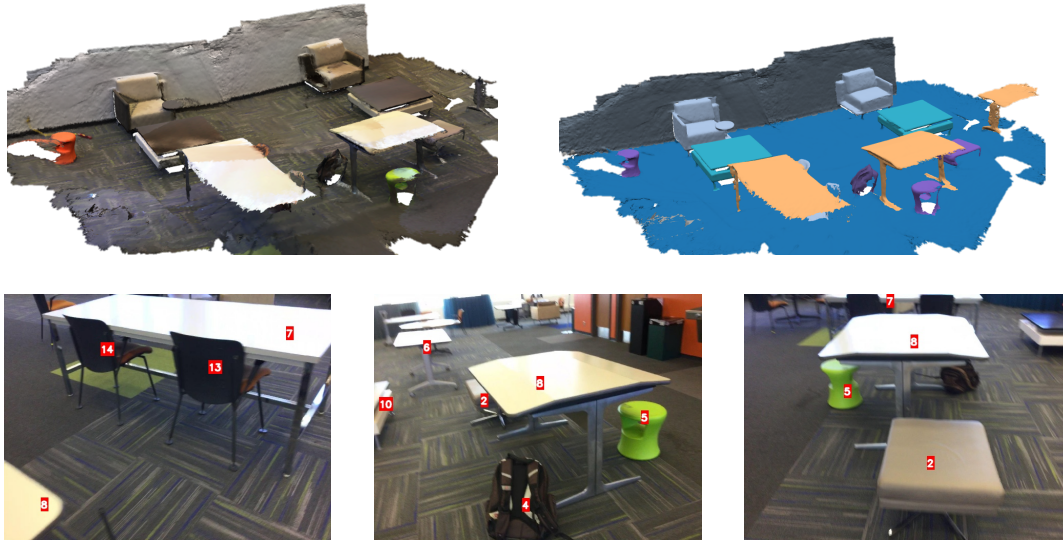
E.2 TwoStage Baseline Agent

The TwoStage agent uses the same prompt as the End2End agent, but uses an additional prompt (shown in Figure 9) to extract the target object type from a given dialogue, which is used to obtain a more focused selection of images to include in the subsequent VLM prompt.

E.3 Modular Agent

For the proposed modular grounding method, we provide the following prompts:

- Dialogue analysis:
 - User answer analysis (Agent): Fig. 10
 - Agent question analysis (User): Fig. 12
- Reference resolution:
 - Object reference extraction: Fig. 11
 - Relevant object types extraction: Fig. 13
 - Information specification: Fig. 14
 - Target object prediction: Fig. 15
- Captioning:
 - Object description generation (User): Fig. 16
 - Ambiguous object description (User): Fig. 17
 - Closed clarification questions (Agent): Fig. 18
- Open clarification question (Agent): Fig. 19
- Answer formulation (User): Fig. 20



Objects: 0. chair, 1. chair, 2. seat, 3. floor, 4. backpack, 5. seat, 6. *table*, 7. *table*, 8. table, 9. coffee table, 10. coffee table, 11. wall, 12. wall, 13. chair, 14. chair, 15. seat

Agent utterance	User utterance	Agent prediction
“Which object did you have in mind?”	“The table”	[6, 7, 8]
“I see three tables, which one do you mean?”	“the one next to a stool”	[8]

Figure 7: Example scenario, showing two views of the point cloud, three images from the real world scene, annotated with object IDs, the list of objects known to the agent, the target object underlined and the distractors italicised, and a dialogue in which the agent interactively identifies the target object (8) known to the user.

You are a helpful assistant specialized in 3D visual grounding.

I will provide a dialogue between an user and an agent, in which the user attempts to locate an object within a scene with the help of the agent. Your task is to identify the target object's ID from a list of objects, based on the user's utterances, the images of the scene, and a top-down view of the scene.

If more than one candidate object likely satisfies the conditions described by the user, include ALL such candidate objects in your response. In that case, also generate a clarification question that clearly distinguishes between the candidates.

Do NOT include object IDs in the clarification question!

Respond ONLY with a JSON object in the following format:

```
{
  "explanation": "Briefly explain why each object ID is or is not identified",
  "object": "Expression of the target object mentioned by the user"
  "object_ids": [<object_id_1>, <object_id_2>, ...],
  "clarification_question": "A question to help resolve the ambiguity between the candidates"
}
```

Now, based on the dialogue, the object list, and the images below, provide your response.

Figure 8: End-to-end baseline prompt for Agent.

You are a helpful assistant that analyzes dialogue transcripts.

You will be given a list of objects in a 3D scene and a dialogue between a user and an agent, in which the user refers to one of these objects and the agent tries to identify that object. Your task is to determine the object class of that target object.

Respond ONLY with one of the object classes in the object list.

Examples:

1.
Objects: 1. table, 2. chair, 3. table, 4. door

Dialogue:
Agent: Which object do you have in mind?
User: It is a table.

Output: table

2.
Objects: 1. chair, 2. chair, 3. lamp, 4. table, 5. chair

Dialogue:
Agent: Which object do you have in mind?
User: A chair.
Agent: I can see three chairs. Which one do you mean?
User: The brown chair next to the table.

Output: chair

3.
Objects: 1. desk, 2. bed, 3. chair, 4. door, 5. backpack, 6. backpack

Dialogue:
Agent: Which object do you have in mind?
User: It is the backpack that is under the bed.

Output: backpack

Now, based on the object list and dialogue below, provide your response.

Figure 9: Prompt for target object type extraction.

You are a helpful assistant specialized in analyzing dialogues.

I will provide a dialogue between an user and an agent, in which the user attempts to locate an object within a scene with the help of the agent. Your task is to predict the next action the agent should take.

Possible Actions:

1. Conclude the dialogue and predict the target object's ID.
2. Perform reference resolution based on the user's response.

Respond ONLY with a JSON object in the following format:

```
{
  "action": 1 or 2,
  "target_object_id": "<object_id>" or null
}
```

Examples:

1.
Dialogue:
- User: Find the brown chair next to the desk.

Output:

```
{
  "action": 2,
  "target_object_id": null
}
```

2.
Dialogue:
- User: Look for the couch in the corner of the room.
- Agent: I see two couches in the corner of the room. Do you mean the grey one or the brown one? (the grey one: 2, the brown one: 5)
- User: It's the brown one.

Output:

```
{
  "action": 1,
  "target_object_id": 5
}
```

3.
Dialogue:
- User: Your target is the orange cup.
- Agent: I found two orange cups. Do you mean the cup on the table or the cup on the shelf? (the cup on the table: 13, the cup on the shelf: 17)
- User: Neither, it is the red one next to the microwave.

Output:

```
{
  "action": 2,
  "target_object_id": null
}
```

Figure 10: Dialogue analysis prompt for Agent (1/2).

4.
Dialogue:
- User: Look at the window.
- Agent: I can see three windows, which window do you mean?
- User: The smallest one on the left.

Output:

```
{
  "action": 2,
  "target_object_id": null
}
```

Now, based on the dialogue below, provide your response.

Figure 10: Dialogue analysis prompt for Agent (2/2).

You are a helpful assistant specialized in 3D visual grounding.

I will provide a dialogue between an user and an agent, in which the user attempts to locate an object within a scene with the help of the agent. Your task is to extract the target object(s) referred by the agent/user.

Respond ONLY with a Python list of target objects.

Examples:

1.
Dialogue:
- User: Find the brown chair next to the desk.

Output:
["the brown chair next to the desk"]

2.
Dialogue:
- User: Look for the couch in the corner of the room.
- Agent: I see two couches in the corner of the room. Do you mean the grey one or the brown one?

Output:
["the grey couch in the corner of the room", "the grey couch in the corner of the room"]

3.
Dialogue:
- User: Your target is the orange cup.
- Agent: I found two orange cups. Do you mean the cup on the table or the cup on the shelf?
- User: Neither, it is the red one next to the microwave.

Output:
["the red cup next to the microwave"]

4.
Dialogue:
- User: Look at the window.
- Agent: I can see three windows, which window do you mean?
- User: The smallest one on the left.

Output:
["the smallest window on the left"]

Now, based on the dialogue below, provide your response.

Figure 11: Object reference extraction prompt for Agent.

You are a helpful assistant specialized in analyzing dialogues.

I will provide a dialogue between a user and an agent, in which the agent attempts to identify a object within a scene that is only known to the user. Your task is to identify the type of the agent's most recent clarification question.

Question Types:
1. Open question
2. Closed question

Respond ONLY with 1 or 2.

Examples:

1.
Dialogue:
- Agent: Which object do you have in mind?

Output:
1

2.
Dialogue:
- Agent: Which object do you have in mind?
- User: the window.
- Agent: I can see three windows, which window do you mean?

Output:
1

3.
Dialogue:
- Agent: Which object do you have in mind?
- User: the couch in the corner of the room.
- Agent: I see two couches in the corner of the room. Do you mean the grey one or the brown one?

Output:
2

4.
Dialogue:
- Agent: Which object do you have in mind?
- User: Find the orange cup.
- Agent: I see two orange cups. One is on the dining table, and the other one is on the shelf, which one do you mean?

Output:
2

Now, based on the dialogue below, provide your response.

Figure 12: Dialogue analysis for User.

You are a helpful assistant specialized in 3D visual grounding.

I will provide a list of object classes in a scene. Your task is to identify those relevant to an utterance.

Respond ONLY with a Python list of selected object classes.

Now, based on the utterance and the object list below, provide your response.

Figure 13: Prompt.

You are a helpful assistant specialized in 3D visual grounding.

I will provide an utterance that refers to an object in a scene. Your task is to identify the target object's ID from a list of objects based on the utterance.

You should determine the relevant information and the associated object IDs that support your decision.

Available Information:

- (a) Object Image: Provides visual details (e.g., color, shape, material, state, etc.).
- (b) Size: Indicates whether an object is big or small.
- (c) Dimensions: Helps differentiate objects as long/short, wide/narrow, or tall/short.
- (d) Distance Between Objects: Indicates whether objects are near or far.
- (e) Multi-Object Image: Shows spatial relationships between objects (e.g. on, above, under, beside, left/right, between, front, behind, etc.).
- (f) Top-Down Scene View: Provides scene layout (e.g. wall, corner or middle of the room).

Respond ONLY with a JSON object in the following format:

```
{
  "(a-f)": {
    "object_ids": [<object_id_1>, <object_id_2>, ...],
    "usage": "Use <information_item> to ..."
  },
  ...
}
```

Now, based on the utterance and the object list below, provide your response.

Figure 14: Information specification prompt for Agent.

You are a helpful assistant specialized in 3D visual grounding.

I will provide an utterance that refers to an object in a scene. Your task is to identify the target object's ID from a list of objects, based on the utterance and the provided information.

Important Instruction

- There may be more than one candidate object that likely satisfies the conditions described in the utterance.
- Include ALL such candidate objects in your response.
- If no candidate object satisfies the conditions, return an empty list.

Respond ONLY with a JSON object in the following format:

```
{
  "explanation": "Briefly explain why each object ID is or is not identified, referring to the provided information",
  "target_object_ids": [<object_id_1>, <object_id_2>, ...]
}
```

Now, based on the utterance, the object list, and the information below, provide your response.

Figure 15: Target object prediction prompt for Agent.

You are a helpful assistant specialized in generating referring expressions for 3D scenes.

I will provide a target object and distractor objects within a scene that all satisfy an utterance. Your task is to generate a referring expression that clearly identifies the target object, distinguishing it from the distractor objects.

You will also have access to other objects in the scene, which can be used as anchors to help refer to the target object.

Available Information:

- (a) Object Image: Provides visual details (e.g., color, shape, material, state, etc.).
- (b) Size: Indicates whether an object is big or small.
- (c) Dimensions: Helps differentiate objects as long/short, wide/narrow, or tall/short.
- (d) Distance Between Objects: Indicates whether objects are near or far.
- (e) Multi-Object Image: Shows spatial relationships between objects (e.g. on, above, under, beside, left/right, between, front, behind, etc.).
- (f) Top-Down Scene View: Provides scene layout (e.g. wall, corner or middle of the room).

Important Instruction:

- NEVER include object IDs in the expression!
- Do NOT use unlisted anchor objects.
- Do NOT repeat information already existing in the utterance.
- Use as little information as possible.
- Use as few anchor objects as possible.
- Do NOT use ambiguous anchor objects.
- When using (b), (c), or (d), ensure the differences of these numbers between the objects are VERY substantial.
- If using left/right orientation, you MUST include a clear anchor object for reference.

Respond ONLY with a JSON object in the following format:

```
{
  "explanation": "Explain why the information is selected (by referring to numbers
  if used)",
  "expression": "referring expression to the target object"
}
```

Examples:

1.

Utterance:

Find the chair.

> Output:

```
{
  "explanation": "...",
  "expression": "It is the brown chair next to the desk."
}
```

2.

Utterance:

Find the cup on the dining table.

> Output:

```
{
  "explanation": "...",
  "expression": "It's the largest one."
}
```

Figure 16: Object description generation prompt (1/2).

```
3.
Utterance:
Find the cabinet by the door.

> Output:
{
  "explanation": "...",
  "expression": "It's the one by the open door."
}

4.
Utterance:
Find the bookshelf.

> Output
{
  "explanation": "...",
  "expression": "It is the bookshelf on the left of the window."
}

5.
Utterance:
Find the grey trash can.

> Output:
{
  "explanation": "...",
  "expression": "It's the one with a white bag."
}

6.
Utterance:
Find the desk with a monitor on it.

> Output:
{
  "explanation": "...",
  "expression": "It's the one in the middle of the room."
}

Now, based on the utterance, the target object, the distractor objects, other objects,
and the information below, generate your referring expression.
```

Figure 16: Object description generation prompt (2/2).

You are a helpful assistant specialized in generating referring expressions for 3D scenes.

I will provide multiple objects within a scene. Your task is to generate an ambiguous referring expression that applies to all of these objects.

You will also have access to other objects in the scene, which can be used as anchors to help refer to the objects.

Available Information:

- (a) Object Image: Provides visual details (e.g., color, shape, material, state, etc.).
- (b) Size: Indicates whether an object is big or small.
- (c) Dimensions: Helps differentiate objects as long/short, wide/narrow, or tall/short.
- (d) Distance Between Objects: Indicates whether objects are near or far.
- (e) Multi-Object Image: Shows spatial relationships between objects (e.g. on, above, under, beside, left/right, between, front, behind, etc.).
- (f) Top-Down Scene View: Provides scene layout (e.g. wall, corner or middle of the room).

Important Instruction:

- NEVER include object IDs in the expression!
- Do NOT use unlisted anchor objects.
- Ambiguity can arise from either the target objects or the anchor objects.
- When using (b), (c), or (d), ensure the differences of these numbers between the objects are VERY small.

Respond ONLY with a JSON object in the following format:

```
{
  "explanation": "Explain the ambiguity in the expression (by referring to numbers if used)",
  "expression": "referring expression to the objects"
}
```

Examples:

1.


```
{
        "explanation": "...",
        "expression": "It is the brown chair."
      }
```
2.


```
{
        "explanation": "...",
        "expression": "Look at the small cup."
      }
```
3.


```
{
        "explanation": "...",
        "expression": "Find the cabinet by the door."
      }
```
4.


```
{
        "explanation": "...",
        "expression": "Your target is the window above the kitchen counter."
      }
```
5.


```
{
        "explanation": "...",
        "expression": "It's the grey trash can with a white bag."
      }
```

Figure 17: Ambiguous object description generation prompt (1/2).

```
6.
{
  "explanation": "...",
  "expression": "Find the desk in the corner of the room."
}
```

Now, based on the ambiguous objects, other objects, and the information below, generate your referring expression.

Figure 17: Ambiguous object description generation prompt (2/2).

You are a helpful assistant specialized in generating clarification questions for 3D scenes.

I will provide two objects within a scene that both satisfy an utterance. Your task is to generate a clarification question that clearly distinguishes between them based on the provided information.

You will also have access to other objects in the scene, which can be used as anchors to help refer to the objects to be clarified.

Available Information:

- (a) Object Image: Provides visual details (e.g., color, shape, material, state, etc.).
- (b) Size: Indicates whether an object is big or small.
- (c) Dimensions: Helps differentiate objects as long/short, wide/narrow, or tall/short.
- (d) Distance Between Objects: Indicates whether objects are near or far.
- (e) Multi-Object Image: Shows spatial relationships between objects (e.g. on, above, under, beside, left/right, between, front, behind, etc.).
- (f) Top-Down Scene View: Provides scene layout (e.g. wall, corner or middle of the room).

You can generate either:

- > a yes-no question by only referring to one of the candidate objects, or
- > a two-option question by referring to both candidate objects, whichever is more natural.

Important Instruction:

- NEVER include object IDs in the question!
- Do NOT use unlisted anchor objects.
- Use as little information as possible.
- Use as few anchor objects as possible.
- Do NOT use ambiguous anchor objects.
- When using (b), (c), or (d), ensure the differences of these numbers between the objects are VERY substantial.
- If using left/right orientation, you MUST include a clear anchor object for reference.

Respond ONLY with a JSON object in the following format:

```
{
  "explanation": "Explain why the information is selected (by referring to numbers if used)",
  "<object_1_id>": "referring expression of the object",
  "<object_2_id>": "referring expression of the object",
  "question": "clarification question"
}
```

Examples:

1.
 - Utterance: Find the trash can.

> Output:

```
{
  "explanation": "...",
  "4": "the one by the door",
  "6": "",
  "question": "I found two trash cans. Do you mean the one by the door?"
}
```

Figure 18: Closed clarification question generation prompt.

```

2.
Utterance:
Find the window above the kitchen counter.

> Output:
{
  "explanation": "...",
  "23": "the bigger one",
  "24": "the smaller one",
  "question": "I can find two windows above the kitchen counter, a bigger one and a
  smaller one. Which one do you mean?"
}

3.
Utterance:
Find the wooden table next to the couch.

> Output:
{
  "explanation": "...",
  "23": "the one next to the black couch",
  "24": "the one next to the red couch",
  "question": "I can see two wooden tables next to a couch. Do you mean the one next
  to the black couch or the one next to the red couch?"
}

Further Question Examples:
4. <ambiguity summarization>. Do you mean the one near the open door or the one near
the closed door?
5. <ambiguity summarization>. Do you mean the taller one?
6. <ambiguity summarization>. Do you mean the one on the desk?
7. <ambiguity summarization>. Do you mean the one on the left of the window or the one
on right?
8. <ambiguity summarization>. Do you mean the one in the middle of the room or the one
in the corner?

Now, based on the utterance, the objects to clarify, other objects, and the information
below, generate your clarification question.

```

Figure 18: Closed clarification question generation prompt.

```

You are a helpful assistant specialized in generating clarification questions for
3D scenes.

I will provide a result of reference resolution, which includes an utterance describing
the target object and multiple candidate object IDs. Your task is to generate a question
that asks for clarification about the intended target object.

Examples:
1.
Result:
{
  "object": "Find the trash can.",
  "object_ids": [3, 6, 12]
}

> Output:
I can see three trash cans, which one do you mean?

2.
Utterance:
Result:
{
  "object": "Find the window above the kitchen counter.",
  "object_ids": [21, 22, 26, 28]
}

> Output:
I found four windows above the kitchen counter, which window do you mean?

3.
Utterance:
{
  "object": "Find the wooden table next to the couch.",
  "object_ids": []
}

> Output:
I can't find any wooden tables next to the couch, which one do you mean?

Now, based on the result below, generate your clarification question.

```

Figure 19: Open clarification question generation prompt for Agent.

You are a helpful assistant specialized in generating responses to clarification questions.

I will provide a dialogue between a user and an agent, in which the agent attempts to identify a object within a scene that is only known to the user. Your task is to formulate an answer to the agent's clarification question, based on the reference resolution result and the target object's ID in mind.

Examples:

1.
Dialogue:
- User: Look for the couch in the corner of the room.
- Agent: I see two couches in the corner of the room. Do you mean the grey one or the brown one?

Result:
{
 "the grey couch in the corner of the room": 12,
 "the brown couch in the corner of the room": 16
}

Target Object ID:
12

Output:
It is the grey one.

2.
Dialogue:
- User: Your target is the orange cup.
- Agent: I found two orange cups. Do you mean the cup on the table or the cup on the shelf?

Result:
{
 "the orange cup on the table": 3,
 "the orange cup on the shelf": 7,
}

Target Object ID:
7

Output:
The cup on the shelf.

Now, based on the dialogue, the reference resolution result, and the target object's ID below, generate your answer.

Figure 20: Answer formulation prompt for User.